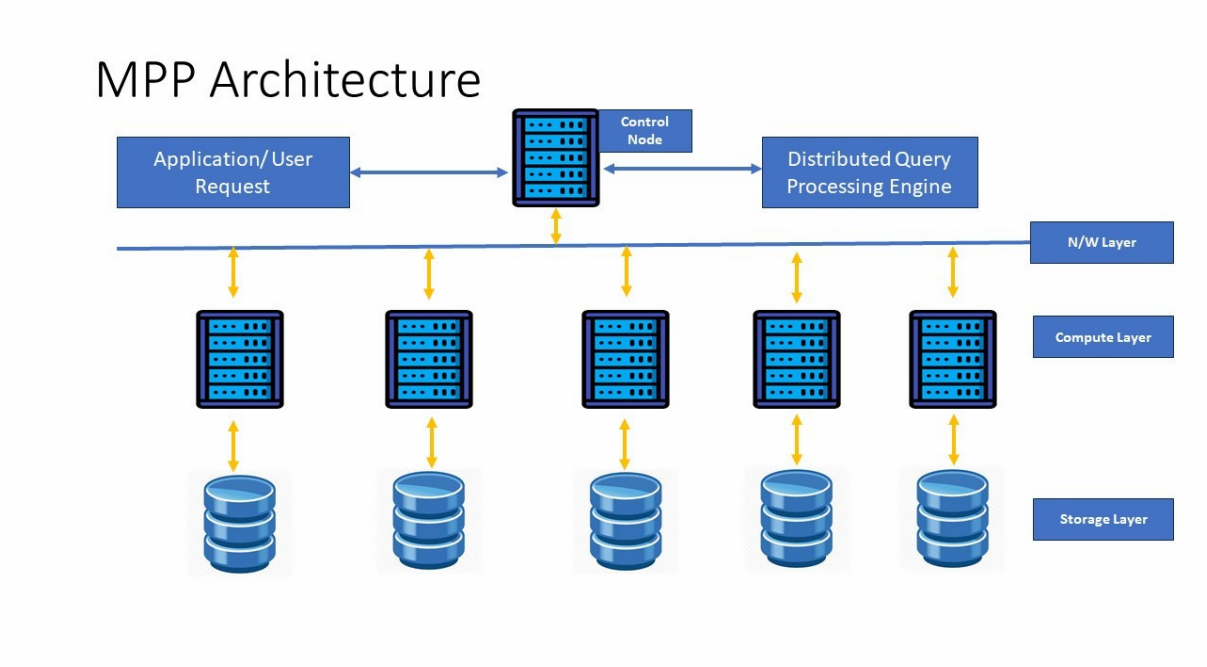

What is Massively Parallel Processing (MPP)?

An MPP database is a database that is optimized to be processed in parallel for many operations to be performed by many processing units at a time.

MPP (massively parallel processing) is the coordinated processing of a program by multiple processors working on different parts of the program. Each processor has its operating system and memory.

In straightforward terms, we are trying to divide and conquer here.

An MPP database or data warehouse partitions both data and computing power among several nodes (servers)

In most technologies, there is a designated leader node that delegates the work, and worker nodes that carry out the tasks.

Important features of MPP Architecture

a) MPP supports shared-nothing Architecture

b) In MPP each processor works on a different part of the task.

c) Each processor has its own set of disks

d) Each node is responsible for processing only the rows on its own disk

e) Scalability is easy by just adding nodes- Horizontal Scaling

For example when there are multiple concurrent users and their queries fighting for resources, we can increase the no. of compute nodes and spread the workload across more resources. This can significantly reduce the data processing times.

f) Usually comes with huge compression ability

g) MPP processors communicate with each other a messaging interface

h) In MPP each processor uses its own operating system (OS) and memory.

Advantages of MPP Architecture

a) Performance: The speed of computation grows linearly. The more nodes you have, the faster it is to perform aggregations and computations on the entire dataset.

b) Scalability: We can scale out in an unlimited fashion. By adding more nodes to our architecture we can scale out our MPP database to store and process larger volumes of data.

c) Cost: Since we are adding more nodes, it is easier to handle more data with less expensive hardware

d) No Single point of failure: If one node fails, other nodes can still be functioning and support the database activities while it is being maintained

a) Elasticity: If I want to add more nodes to the database, it is easy to do so without making the entire cluster unavailable.